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Naaravali

Smart Campus Waste Management Platform

"Turning waste into a cleaner, greener tomorrow."

Team Name: Naaravali

Team Members: Krish Bansal | Manthan Soni | Tejas A

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THE PROBLEM

What challenge are you solving?

Improper Waste Segregation

Mixed waste ends up in landfills, reducing recycling potential.

Inefficient Collection

Fixed routes and no real-time monitoring lead to overflow and excess travel.

Low Recycling Rates

Valuable materials are not recovered due to lack of awareness and infrastructure.

Lack of Data & Visibility

No real-time insights on waste generation and segregation performance.



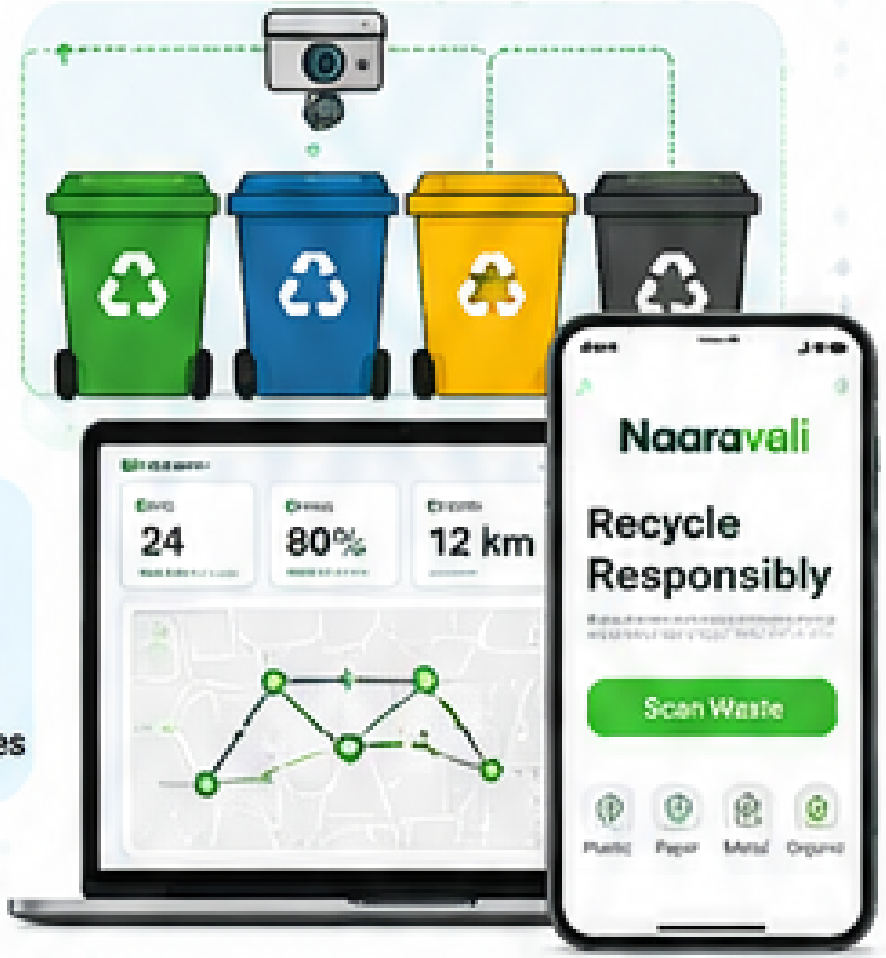
Result:
More landfill waste, higher costs, and a larger environmental impact.

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THE IDEA

Naaravali – Smart Campus Waste Management Platform

An integrated, technology-driven solution that uses AI, IoT, and data analytics to make waste segregation, collection, and recycling smarter, more efficient, and more sustainable in campuses and communities.



Smarter Segregation

Efficient Collection

Higher Recycling Rates

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KEY FEATURES

AI-Powered Waste Identification

- Image-based waste classification
- Smart bin guidance
- Detects incorrect segregation

Smart Collection & Route Optimization

- Real-time bin fill-level monitoring
- Optimized collection routes
- Predictive analytics for high waste areas

Recycling & Resource Recovery

- Connects waste generators with recyclers
- Food-waste composting support
- Tracking of recyclable materials

Data-Driven Monitoring & Insights

- Real-time dashboards
- Waste generation trends
- Segregation & recycling rates
- Forecasting future waste patterns

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INNOVATION

"what makes it different?"

THE DIFFERENTIATOR

AI + IoT Integration

Combines computer vision with smart bins for real-time, accurate waste identification and monitoring.

End-to-End Approach

Covers segregation, collection, recycling, and data monitoring in a single, unified platform.

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Affordable & Scalable

Designed for real-world implementation in campuses and communities with low-cost solutions.

Data-Driven Impact

Provides measurable insights to improve recycling rates and reduce landfill waste over time.

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TECH STACK

Python
AI/ML, Data Analysis, Backend

React
Web Application Frontend

Node.js
Backend APIs

MongoDB
Database

OpenCV
Computer Vision & Image Classification

YOLO
Waste Detection Model

IoT Sensors
Bin Fill-Level Monitoring

Google Maps API
Route Optimization & GIS

Render
Deployment

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IMPACT & SCALABILILTY

Environmental Impact
Higher recycling rates, reduced landfill waste, and cleaner campuses.

Social Impact
Increased awareness and participation in waste segregation.

Economic Impact
Lower collection costs and potential revenue from recyclable materials.

Scalability
Modular, affordable, and replicable solution for multiple buildings, campuses, and communities.



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FROM IDEA TO IMPACT

"Smarter waste management today for **cleaner campuses** and **a greener tomorrow.**"

ONE POWERFUL STATEMENT